Article title: Deep Learning for Skin Cancer Image Analysis: A Systematic Literature Review
Journal name: Informatics in Medicine Unlocked
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LLM Data Extraction Prompt

Role:

You are an expert academic researcher conducting a Systematic Literature Review (SLR) on Deep Learning for Skin Lesion Analysis and Dermatological Disease Diagnosis.

Context & Research Questions (RQs):

Your extraction must be targeted to answer the following specific Research Questions. The scope focuses on skin cancer image analysis, encompassing malignant and precancerous skin lesions (e.g., Melanoma, Basal Cell Carcinoma, Squamous Cell Carcinoma) and their benign mimics (e.g., Melanocytic Nevus, Seborrheic Keratosis). Read the paper with these questions in mind:

RQ1 (Datasets): What datasets are used? (Public or Private?)
RQ2 (Preprocessing): What image-level preprocessing and augmentation strategies are applied BEFORE the model? (e.g., Hair removal, resizing, color normalization. NOT feature extraction).
RQ3 (Lesion Seg): Methodologies for lesion boundary segmentation.
RQ4 (Attribute Seg): Methodologies for specific attribute segmentation (e.g., pigment network, globules).
RQ5 (Detection): Binary classification tasks (e.g., Malignant vs. Benign, Healthy vs. Disease). Specify the classes.
RQ6 (Multi-Class): Multi-class classification tasks (e.g., Melanoma vs. BCC vs. Nevus).
RQ7 (Models): Prevalent architectures/backbones used.
RQ8 (Benchmarks): Metrics and SOTA performance reported.
RQ9 (Challenges): Limitations reported by authors.
RQ10 (Future): Future research directions.

Instructions:

Analyze the attached PDF full text.
Extract precise data for the columns listed below.
If a specific task is NOT performed, write “N/A”.

CRITICAL FORMATTING RULE: Output **ONLY** the raw data row separated by pipes (|).

Data Columns to Extract (In this exact order):

Title
Year
RQ1_Datasets (e.g., “HAM10000”, “Private Hospital Data”)
RQ1_Image_Count (Total images)
RQ1_Class_Balance (Balanced? Yes/No/Method used)
RQ2_Preprocessing (e.g., “Hair Removal, Resizing” or “N/A”)
RQ2_Augmentation (e.g., “Rotation, Flip, GANs”)
RQ3_RQ4_Seg_Scope (Values: “Lesion Boundary”, “Attributes”, “Both”, or “N/A”)
RQ5_RQ6_Class_Scope (Values: “Binary (Classes)”, “Multi-class (Classes)”, “Both”, or “N/A”)
RQ7_Model_Architecture (e.g., “ResNet50”, “U-Net”, “ViT”)
RQ7_Novelty (e.g., “Standard”, “New Architecture”, “Ensemble”, “Transfer Learning”)
RQ7_Transfer_Learning (Yes [Source] / No)
RQ8_Metrics_Used (e.g., “ACC, Dice, AUC”)
RQ8_Best_Result (e.g., “ACC: 98.5%”)
RQ8_SOTA_Compare (Yes/No)
RQ9_Limitations (Brief summary)
RQ10_Future_Work (Brief summary)

Output Template:

{Content_1} | {Content_2} | {Content_3} | {Content_4} | {Content_5} | {Content_6} | {Content_7} | {Content_8} | {Content_9} | {Content_10} | {Content_11} | {Content_12} | {Content_13} | {Content_14} | {Content_15} | {Content_16} | {Content_17}